

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (AIML) (Credit System) Degree Examinations**

January 2023

20AM505 - BIG DATA ANALYTICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the 3V's of Big Data with its representation. How 3V is modified into 4V? Elaborate it clearly.	10	L2	1	1
	b) Summarize between Data Warehouse and Data Mining. Show the components of typical Data Warehouse.	10	L2	2	1
2.	a) i) How Big Data types are classified. Explain. ii) Illustrate how Data Warehouse and Big Data environments come together in an integrated and complementary way.	6+4	L2	1	1
	b) Define Analytics. Explain three types of data analytics.	10	L2	2	1
3.	a) Relate in detail the Industrial examples of Big data. (any 5)	10	L2	1	1
	b) i) Illustrate any 5 difference between RDBMS and Hadoop. ii) Explain YARN map reducing components of resource manager.	5+5	L2	2	1
Unit – II					
4.	a) Explain in detail categories of NO SQL databases.	10	L2	3	1
	b) What is MongoDB? Why it is required? Explain in detail.	10	L2	4	1
5.	a) Explain in detail the physical architecture of HBase.	10	L2	4	1
	b) How HBase can be represented in key/value and multi-map structure? Explain in detail.	10	L2	3	1
6.	a) Perform following operations with reference to CQLSH. i) To describe column languages as "Known Languages" in student_info ii) To sort the rows/records of the project_details column in ascending order of the project_name. iii) List and Explain the key components of Cassandra.	5 5 5	L2 L2 L2	4 3 4	2 1 1
	b) Explain with diagram the working of map reduce.	10	L2	4	1
Unit – III					
7.	a) Illustrate with diagram the workflow between Hive and Hadoop.	10	L2	5	1
	b) Write a note on i) Hive Query Language ii) Data types used in hive.	10	L2	5	1
8.	a) With suitable diagram explain the main building blocks of Hive.	10	L2	5	1
	b) Write a note on (i) Pig and its components (ii) Models of execution in Pig.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
